

SELECTION METHODS

Choosing Which Agents Survive

Overview

Selection determines which agents survive to breed. The right selection method balances exploitation (keeping the best) with exploration (maintaining diversity).

Methods

Tournament Selection

Random subsets compete; winners advance.

Process:

1. Select k random agents (tournament size)
2. Choose the fittest from the tournament
3. Repeat until population filled

Pressure: Adjustable via tournament size

Roulette Wheel Selection

Probability proportional to fitness.

Process:

1. Calculate total fitness
2. Assign probability = $\text{fitness} / \text{total}$
3. Spin the wheel to select

Pressure: Moderate; fit agents more likely but not guaranteed

Rank Selection

Selection based on rank, not absolute fitness.

Process:

1. Sort agents by fitness
2. Assign selection probability by rank
3. Select based on rank probability

Pressure: Consistent; avoids domination by outliers

Elitism

Best agents always survive.

Process:

- 1. Keep top N agents unconditionally
- 2. Fill remaining slots with other methods

Pressure: High; guarantees best traits preserved

Comparison

Method	Pressure	Diversity	Best For
Tournament	Medium	Medium	General use
Roulette	Low	High	Exploration
Rank	Medium	Medium	Avoiding premature convergence
Elitism	High	Low	Preserving breakthroughs

Recommendation

Default: Tournament (k=3) + Elitism (top 2)

This balances exploration and exploitation for most use cases.

SELECTION | Survival of the Fittest